



Estimating (EST)

EST Overview

Required PA Information

Intent

Estimates the size, effort, duration, and cost of the work and resources needed to develop, acquire, or deliver the solution.

Value

Provides a basis for making commitments, planning, and reducing uncertainty, which allows for early corrective actions and increases the likelihood of meeting objectives.

Additional Required PA Information

If Safety is included in the selected view: Include safety-related requirements, activities, tasks, risks, and assumptions when formulating estimates, e.g., regulations, physical safety constraints, required safety training, accommodations for Personal Protective Equipment (PPE) and physical distancing, safety audits and certifications, and schedule and cost buffers for safety incident handling.

If Security is included in the selected view: Include security-related requirements, activities, tasks, risks, and assumptions when formulating estimates, e.g., security clearance paperwork, background checks, regulations, system access, required security training, and schedule and cost buffers for security incident handling.

Explanatory PA Information

Practice Summary

Level 1

EST 1.1 Develop high-level estimates to perform the work.

Level 2

EST 2.1 Develop, keep updated, and use the scope of what is being estimated.

EST 2.2 Develop and keep updated estimates for the size of the solution.

EST 2.3 Based on size estimates, develop and record effort, duration, and cost estimates and their rationale for the solution.

Level 3

EST 3.1 Develop and keep updated a recorded estimation method.

EST 3.2 Use the organizational measurement repository and process assets for estimating work.

Additional PA Explanatory Information

Estimation provides the basis for making commitments. An estimate takes into consideration the scope, size, and complexity of the work. Base the estimate on the available information. Record any uncertainty as a risk.

Uncertainty or risk in a commitment can be handled by:

- Providing an initial commitment with an understanding that the initial commitment may be changed if the scope changes
- Defining milestones to refine an initial commitment range to produce a final commitment after more investigation
- Committing to the known parts of the project, and committing to the remainder of the work after further investigation and definition

Historical data describing the relationship between measured size and resources such as effort, cost, and schedule should be used when planning future work. A good understanding of historical data is critical to successful estimating. Use historical data when planning future work and to calibrate estimation formulas and models. Record qualitative information such as context, methods, tools and techniques used, and lessons learned from past projects.

Estimate and track several aspects of the work to realize value. For example, based on [Table EST-1: Example Tracking Information](#), how complete is the work?

Table EST-1: Example Tracking Information

Aspect	Percent Completed
Effort	60%
Duration	50%
Cost	75%

This question cannot be answered if only effort is tracked. To get a complete picture of the status, estimate and track the other aspects of the project. If all aspects are not estimated and tracked, it could lead to an incomplete or misleading understanding of the work status. The numbers in the table indicate a potential problem and should trigger an investigation to determine the real status of the project. There are other aspects of the work, such as complexity, that may affect the answer. For example, the work status may only be 10% complete because the project front loaded the highly complex activities or components. Understanding the aspects and their relationships provides a more comprehensive estimate. The rationale of the estimate should be based on historical data, rather than solely on the experience and knowledge of the estimator.

Related Practice Areas

Managing Performance and Measurement (MPM)

Planning (PLAN)

Requirements Development and Management (RDM)

Risk and Opportunity Management (RSK)

Context Specific

Agile Development

Context Tag:	Agile Development
Context:	Integrate agile techniques and ceremonies with other processes.

Agile development teams estimate the work during backlog grooming and Sprint planning sessions:

- Estimates for backlog items are typically a rough order of magnitude
- Some agile development teams develop a comprehensive estimate of work during release planning for a set of stories or epics
- Estimates for Sprints are typically more refined, allowing the team to understand commitments

Agile estimation typically includes:

- **Size:** During backlog review, assign backlog items, such as requirements and user stories, a relative size using story points. The transformation of user stories into story points considers the size and complexity of the solution. In addition to story points, agile development teams may use t-shirt size estimation (small, medium, large, or extra-large). Often requirements are converted into user stories before estimating is performed. Complex needs or requirements may be transformed into an epic, which is typically a large user story that can span more than one Sprint. If the epic spans more than one Sprint, it is typically broken into smaller user stories.
- **Tasks and Effort:** During Sprint estimating and Sprint planning, agile development teams and the product owner collaborate to select user stories off the backlog based on the priority of the product owner. The team then estimates these user stories using relative sizing techniques such as story points or t-shirt sizing. Using the team's known velocity as a guide, prioritized stories are accepted by all stakeholders into the Sprint. Some agile teams estimate the task effort in hours for each story based on historical data or other effort estimation technique.
 - Determine how many user stories can be committed to the Sprint when team velocity is known (story points completed per Sprint)
 - Estimate effort at the task level and use the total to determine the amount of work that can fit into a Sprint based on available capacity
 - Use known velocity numbers to make a first estimate of what can be committed to in a Sprint, and then use task breakdown and effort data to refine and validate the decision
- **Task Assumptions:** Discuss and confirm assumptions during Sprint planning events and review during the retrospective to improve estimates. Record, clarify, and communicate assumptions during these events.

Examples of agile activities with corresponding estimating descriptions are reflected in [Table EST-2: Example Agile Estimating Activities](#).

Table EST-2: Example Agile Estimating Activities

Agile Activities	Description
Release Planning	Identifies sequencing and high-level estimates for all planned epics and stories
Backlog Grooming	Refines stories and acceptance criteria to verify and understand the work to be performed. Sprint velocity is used to estimate time to complete remaining story points.
Sprint Planning	Allocates story points across Sprints
Sprint Review/Demo	Reviews actual results for each Sprint and velocity for the just completed Sprint, and realign sequencing of remaining epics and stories across future Sprints
Retrospective	Identifies consistencies in hitting target Sprint velocity and estimates any unplanned work and improvement opportunities

People

Context Tag:	CMMI-PPL
Context:	Use processes to align workforce performance and business objectives.

A defined method to forecast workforce requirements including estimates of different resource categories, types, and amounts is key to ensuring adequate resources and skills are available when needed to fulfill relevant business needs.

Examples of historical actuals and related estimates to consider include:

- Internal personnel transitions
- Voluntary and involuntary attrition
- Recruitment and outsourcing targets
- Budget and timing for promotions
- Workforce reduction trends and expectations

Level 1

EST 1.1

Required Practice Information

Practice Statement

Develop high-level estimates to perform the work.

Value

Addresses work size, cost, and schedule uncertainties to avoid pursuing work that may result in schedule or budget overruns.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

The high-level estimate is typically:

- A rough draft, top-down estimate (also called a rough order of magnitude estimate)
- Based on identified or recorded assumptions and uncertainty
- Developed quickly
- Based on previous knowledge and experience

Example Activities

Example Activities	Further Explanation
Review needs and assumptions and determine high-level estimates with stakeholders.	

Example Work Products

Example Work Products	Further Explanation
Rough order of magnitude estimate	Includes: • An estimate of the size, complexity, cost, effort, or duration of the solution • Assumptions • Unit of measure

Level 2

EST 2.1

Required Practice Information

Practice Statement

Develop, keep updated, and use the scope of what is being estimated.

Value

Ensures the entire solution is addressed which increases the likelihood of meeting objectives and avoiding rework.

Additional Required Information

The scope should:

- Define the solution to be developed, delivered, or acquired
- Specify the information needed to estimate the size, effort, cost, and duration
- Identify any resources needed, acquired or consumed during the project
- Identify resource capacity and availability needs and constraints
- Establish the work constraints, e.g., what is included and what is not included such as acceptance or test criteria

Explanatory Practice Information

Additional Explanatory Information

Use an initial set of requirements and work objectives to form the basis for establishing the scope. Defining and using the scope can help uncover missing or misunderstood requirements, identify risks or opportunities, and develop more detailed estimates. Inaccurate estimation is often the result of not understanding the scope of the work. Update the scope as the project progresses to address changes.

Example Activities

Example Activities	Further Explanation
Review requirements and objectives with stakeholders to determine scope.	Define the solution to be developed, delivered, or acquired.
Gather information to estimate the size, effort, cost, resources, and duration.	Includes both resource capacity and availability.
Identify constraints, boundaries and limits of scope.	

Example Work Products

Example Work Products	Further Explanation
List of tasks and activities or Work Breakdown Structure (WBS)	To estimate costs more accurately, include identified resources for tasks and durations.
List of needed resources	Includes not just personnel, but also other resources needed to accomplish the work, e.g., facilities, acquired solutions, tools.
Workflow diagram	Visualizes how tasks will flow between resources and what conditions allow the sequence to move forward.

Related Practice Areas

Requirements Development and Management (RDM)

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Consider activities associated with the acquisition strategy when determining scope. For example, a complex project can involve managing multiple supplier agreements with one or more suppliers.

EST 2.2

Required Practice Information

Practice Statement

Develop and keep updated estimates for the size of the solution.

Value

Enables work tracking and timely corrective actions to deliver the solution on time and within budget.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Size is the primary input to many estimation models. Estimation is not a one-time activity that is only performed before or at the start of the project. It is a recurring activity where the estimate is adjusted as new information becomes available throughout the lifecycle of a solution, operations, and maintenance, or for services produced and delivered.

Estimating size provides a consistent basis for estimating effort, duration, and cost. A relative level of difficulty or complexity may be associated with size estimates and is used in the transformation to effort, duration, cost, and quality. For example:

- For services, size could be the type or number of service requests, number of calls received in an hour, or the number of customers desiring a service delivery, etc.
- For software development, size could be the number of objects, the number of components, the number of features, standard or customized function points, the number of requirements, or the number of lines of code, etc.
- For hardware development, size could be the number of connections or connection points, the number of welds, the number of boards, the number of components, or the number of hardware and software integration points, etc.
- For supplier management, size could be the number of requirements, the number of features, the number of items to be acquired, or the number and types of bidders, etc.

Example Activities

Example Activities	Further Explanation
Use applicable methods to estimate the size and complexity of solutions and tasks.	Methods for determining size include: • Analogy • Delphi • 3-point estimation • Parametric estimation The project estimation methods and their use may change over time as the understanding of the relationship of solution characteristics to size improves. Complexity is typically used in the transformation from size to effort, duration, and cost. Complexity may also include qualitative aspects of the solution, such as new vs. legacy.

Example Work Products

Example Work Products	Further Explanation
Size estimate	Typically includes: • Size • Unit of measure

Example Work Products	Further Explanation
	• Rationale or basis for estimate, including assumptions and constraints • Complexity – could be a multiplier of size, or a modifier (such as Hard, Medium, Easy), to consider the potential difficulty of implementing the solution

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Characteristics used to develop estimates include:

- Number and type of services
- Number of service levels
- Volume of service requests

Examples of tasks to develop size estimates for include:

- Service system development and delivery
- Service system monitoring
- Preventative maintenance or repair
- Training
- Incident management and resolution
- Updating equipment and supplies
- Logistical support
- Facilities maintenance
- Transition activities
- Monitoring for and addressing obsolescence
- System disposal

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Examples of characteristics used to estimate include:

- Number of tasks and effort needed to manage the supplier
- Number and complexity of work products
- Size of work products, e.g., pages, screens, inputs/outputs, tickets, lines of code, number of story points, or number and size of deliverables
- Number and complexity of requirements in the supplier request package

- Number and complexity of clauses in the supplier agreement
- Amount and complexity of the work suppliers will address
- Number of potential suppliers
- Prioritized solution features

EST 2.3

Required Practice Information

Practice Statement

Based on size estimates, develop and record effort, duration, and cost estimates and their rationale for the solution.

Value

Enables a better basis for commitments and improves accuracy of the estimates, leading to better decision-making.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

Transform the size estimate into estimates of effort, duration, and cost. Use estimation models, historical data, expert judgment, or a combination of all three. Understanding the size of the solution provides a more accurate basis for determining the effort, duration, and cost for the solution. While performing the work, compare the rationale to actual conditions to identify missing or unnecessary aspects in the original estimate. Identifying missing or unnecessary aspects supports replanning the current work or estimating future work.

Managers and leads typically perform top-down estimation. Bottom-up estimation is typically performed by team members.

Develop and calibrate estimation models using available historical data. To increase confidence, update estimation models as additional data becomes available.

Sometimes, historical data is not available, such as when efforts are unprecedented. Unprecedented efforts are riskier and require more research to develop a basis of estimate. Record rationale for what made the work unique to aid understanding of any assumptions made in the initial planning phases.

Example Activities

Example Activities	Further Explanation
Collect and use historical data to develop, calibrate, or recalibrate models or methods to transform size and complexity into effort, duration, and cost estimates.	To ensure a high level of confidence in the estimate, use multiple models or methods as needed, for example, based on the most important set of tasks and activities. Historical data should include: • Size, cost, effort, and schedule (duration) data from previous completed projects • Appropriate scaling data to account for differing sizes and complexity • Information on factors that influenced the performance and other contextual information, when available, will help determine if past data can be included, excluded, or adjusted Historical data can also be used with analogies, e.g., if a current project is 10% smaller than a similar historical project, use the historical project's results reduced by 10%. There may be instances where historical data is not available or does not apply. In the absence of historical data (for example, no prior history of work similar to current work) external sources like industry data may be used. Models can also be based on other characteristics such as service level, connectivity, complexity, availability, reusability, and structure. Other examples of characteristics include: • Critical competencies and roles needed to perform the work • Needed knowledge, skills, experience, and training • Selected lifecycle model and processes • Travel • Team productivity • Geographic dispersal of work group members • Proximity of customers, end users, and suppliers • Amount of risk • How agreeable or difficult the customer is • Direct labor rates and overhead • Penalties for warranty work • Regulatory requirements or environment • Level of security required for tasks, work products, hardware, software, personnel, and work environment
Describe and record the rationale for the estimates of effort, duration, and cost for the solution.	Recording the rationale provides the context for using historical data for estimating future work.

Example Activities	Further Explanation
Include estimates of supporting infrastructure needs.	The supporting infrastructure includes resources needed to support the project but are not necessarily included in the project itself. Consider the infrastructure resources needed for the work, including: • Contracts • Facilities • Tools • Consumables • Licenses • Travel

Example Work Products

Example Work Products	Further Explanation
Effort estimate	Typically includes: • Effort • Unit of measure (typically hours or days) • Productivity • Context for the effort estimate
Duration estimate	Typically includes: • Duration • Unit of measure (typically hours or days) • Rationale for the duration estimate
Cost estimate	Typically includes: • Cost • Unit of measure, e.g., local currency, contract currency • Rationale for the cost estimate
Estimating rationale	Typically includes: • Description of what is being estimated • Scope • Assumptions and constraints • Comparisons to similar work • Team experience with the technology and domain • Risks • Use of historical data • Tools, techniques, or methods used: ○ Off the shelf tools ○ Internally developed tools ○ Formulas and calculations ○ Models

Context Specific

Services

Context Tag:	CMMI-SVC
Context:	Use processes to deliver, manage, and improve services to meet customer needs.

Service estimates should consider the effort and cost associated with delivering the service. Individual services can have associated workflows or detailed steps that involve points of communication, evaluation, and decision. Consider these lifecycles when estimating the requirements to support the delivery of individual services.

Parameters to consider include:

- Service characteristics
- Service system and service system components
- Delivery environment

When estimating effort and cost, include infrastructure resources that support services. For example:

- Computer workstations
- Power, space, and cooling requirements
- Tools for use by service teams
- Facilities
- Network and communications requirements
- Machinery and equipment
- Support for shift work

Inputs used for estimating effort and cost include:

- Availability of services, by service level, e.g., turnaround time, operational availability ratio, number of calls the help desk should be able to handle per hour
- Level of security required for tasks, work products, hardware, software, personnel, and the work environment
- Service and service system requirements
- Service approach
- Size estimates of work products, tasks, and anticipated changes
- Cost of externally acquired products
- Capability of tools provided
- Capability of manufacturing processes
- Experience of service participants
- Proximity of customers, end users, and suppliers
- Technical approach
- Consumables (resources that the service provider needs to replenish or replace before, during, or after providing a service)

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

The amount of supplier work largely determines the amount of acquirer work required to manage the project and the supplier. Effort for the acquirer includes effort associated with:

- Defining the project and scope
- Developing the solicitation and supplier agreement
- Managing the agreement and technical activities
- Planning, monitoring, and controlling the project and supplier
- Developing and updating acquisition requirements

The project derives detailed estimates for activities performed by the acquirer and its stakeholders. The acquirer should include stakeholder representatives to ensure they have accounted for all technical or service considerations in the estimates. As the work evolves, revise estimates based on changing conditions or requirements.

Additionally, the acquirer needs to estimate the cost and effort for the acquired solutions. Estimates should address effort and cost for supplier management and reporting requirements. The acquirer should review its supplier effort and cost estimates with external individuals to ensure reasonable estimates.

Level 3

EST 3.1

Required Practice Information

Practice Statement

Develop and keep updated a recorded estimation method.

Value

Maximizes consistency and efficiency for developing accurate estimates and increases the likelihood of meeting objectives.

Additional Required Information

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Explanatory Practice Information

Additional Explanatory Information

A defined estimation method is a standard approach using established processes and the best available valid data to estimate the current or future size, effort, cost, and duration of a project based on what is known. Valid estimating data is applicable to the context of the work being estimated. Organizations may have more than one estimation method.

Methods using historical data provide a data driven approach to estimation. Calibrate methods based on actual results vs. historical data and recalibrate methods when conditions, assumptions, processes, or performance change. Use an analysis of estimation accuracy to improve the method.

Some standard methods are described in [Table EST-4: Example Estimation Methods](#).

Table EST-4: Example Estimation Methods

Method	Brief Description
Delphi method	Estimates are developed by a group of Subject Matter Experts (SMEs) where each independently gives their estimates and assumptions to the designated facilitator. The team discusses the differences and re-estimates. This is repeated until the estimates converge. The facilitator then records the final estimate.
Comparative or analogous estimation	Estimates are based directly on past results for similar projects. The estimate is then adjusted for differences in size, complexity, or other factors to reflect current knowledge.
Parametric estimation	Parametric estimates are based on historical data and project parameters and typically use a tool. Note that there are various forms of estimation tools, including: • Mathematical

Method	Brief Description
	• Scenario-based • Simulation Calibrating the estimation tool as it is used can provide additional estimation accuracy.
3-point estimation	Each estimator (note that there may be only one estimator) provides high, low, and most-likely estimates. The facilitator combines them and calculates the resulting value by using the formula: $(\text{high} + (4 * \text{most likely}) + \text{low}) / 6$

Example Activities

Example Activities	Further Explanation
Determine the acceptable estimation methods.	Estimation methods are most accurate when based on historical data and validated before use. The estimation methods should be used consistently for similar activities, projects, domains, etc. Involve SMEs in developing and approving the method. The estimation tools can be built or acquired but should be calibrated with organizational data.
Calibrate and adjust method based on actual results.	One approach to calibrate is to refresh the data periodically or recalibrate data after the most recent use. For example, compare the actuals from the most recent calendar quarter of data to an estimate of that same quarter using the average of the previous three calendar quarters. Based on differences, adjust the method.
Validate method.	The method should be validated by SMEs who have used and understand when to apply the method.

Example Work Products

Example Work Products	Further Explanation
Recorded estimation methods	The process, tools, and data used for the selected estimation methods.

EST 3.2

Required Practice Information

Practice Statement

Use the organizational measurement repository and process assets for estimating work.

Value

Increases estimation precision, accuracy, and consistency enabling better decision-making, a higher likelihood of meeting objectives, and reduced risk.

Additional Required Information

This section left blank for future content.

Explanatory Practice Information

Additional Explanatory Information

Using organizational assets as a basis for estimating leverages the data and experience from previous projects to improve the reliability of estimates for similar work. Select the most appropriate estimation method and use it to produce the estimates.

When using organizational assets consider:

- Historical and validated data from this work or similar work and its context
- Similarities and differences between the current work and work from which historical data will be used
- Rationale used to select the historical data
- Type of work
- Tailoring performed
- Geographic-specific information
- Domain and technology

Examples of data contained in the organization's measurement repository that could be used in estimation may include:

- Size
- Effort
- Cost
- Duration
- Personnel
- Experience
- Response time
- Capacity
- Performance
- Quality
- Context specific information

Example Activities

Example Activities	Further Explanation
Use organizational assets and measures for estimation.	Include selection criteria and rationale for the chosen estimation technique.
Use estimation methods.	
Contribute results and measures to the organization to improve the	Include actual results, contextual information, and identified improvements.

Example Activities	Further Explanation
estimation methods and update organizational assets.	
Analyze organizational data.	Analyze data to better understand: • Variability • Data quality • Mean, median, mode

Example Work Products

Example Work Products	Further Explanation
Work estimates	Include historical data, context, and approved use from the organization.
Updated organizational process assets	Updated organizational process assets may include: • Templates • Best practice examples • Approved methods for use • Guidelines Updates to the organizational measurement repository may include: • Historical estimation data, e.g., actual effort expended, number of function points • Rationale for the estimate, e.g., team skill levels, amount of code reused • Contextual information, e.g., domain, type of work, customer • Updated estimation results

Context Specific

Suppliers

Context Tag:	CMMI-SPM
Context:	Use processes to identify, select, and manage suppliers and their agreements.

Parameters to consider include:

- Acquisition types
- Supplier types
- Supplier on-time performance record
- Supplier performance ratings
- Domain-specific characteristics of the solution